AN APPROACH TO LAVA TUBE DETECTION IN RADAR SOUNDER DATA OF THE MOON

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ABSTRACT

Lunar lava tubes are buried channels that contained thermally insulated lava during the volcanic period of the Moon. Nowadays, they are believed to be empty and thus, identified as potential habitats for humans. In recent years, numerous studies investigated the possible locations of these tubes by taking into account the distribution of gravity anomalies and the volcanic features of the surface. In this paper, we model lava tubes according to their electromagnetic behavior, and we propose a novel approach to locate lava tubes and estimate their physical properties. The method analyzes the subsurface reflections stored in radargrams to extract the desired features automatically. Then, these features and their relationships are processed by a fuzzy rule-based system to detect the presence or absence of lava tubes. The strategy was implemented and successfully tested on simulated radargrams with various surface properties and tunnel dimensions.

Index Terms— Lava Tube, Radar Sounder, Line Detector, Fuzzy Logic, Remote Sensing.

1. INTRODUCTION

During the last few decades, an increasing number of studies underlined the importance of finding an alternative place to the Earth where humankind can live. In this scenario, lava tubes on the Moon are an optimal place, since they are a natural shelter against the threats that are present on the surface, such as cosmic rays, solar particle events and extreme variation of temperature during the day [1]. In the literature, two main approaches exist to detect lunar lava tubes. The first is to analyze gravity data acquired by the GRAIL mission to identify gravity anomaly distributions that are compatible with lava tubes induced mass deficit [2]. The second is to analyze volcanic features on the Moon surface, including pits and rilles [1], which can be interpreted as locally collapsed tubes. Both provide information on the probable locations of lava tubes, but no exact information on their size and depth. The paper proposes an alternative approach based on the analysis of radar sounder data, which contain information on the dielectric and physical characteristics of the subsurface structures, and thus, lava tubes can be quantitatively characterized.

A radar sounder irradiates an electromagnetic wave that is able to penetrate the surface and thus, to propagate in the subsurface. A part of the wave is reflected by each interface and collected by the radar antenna. Interfaces are zones with a strong variation in the dielectric properties and are therefore seen as scatters inside a background material. The coherent collection of these echoes is the radargram, which contains all the information about the subsurface scatters. Two studies mapped terrestrial lava tubes by analyzing ground penetrating radar (GPR) data, the terrestrial version of a radar sounder. Rowell et al. [3] studied the Eastern Snake River Plain in southern Idaho, USA, exploiting apriori knowledge of the tunnels geometries. Miyamoto et al. [4] analyzes the peaks in A-scan of radargrams of Fuji Mount, Japan. The possibility to study lava tubes in the space has been investigated recently. For example, [5] studied the properties of an ad hoc Moon radar sounder mission for lava tube detection.

This paper proposes an unsupervised approach to detect the presence or the absence of lava tubes in radargrams. The approach exploits the peculiar electromagnetic signature of tubes, which is due to the transition rock-void-rock in the subsurface encountered by the propagating wave to define the electromagnetic model of lava tubes. On that model, a method is defined based on two steps: the first extracts linear features from radargrams. The second step consists of a fuzzy system that evaluates the features and their relationships. This approach was implemented and successfully tested by using simulated radargrams.

The structure of the paper is the following: Section 2 describes the geological and electromagnetic model for lava tubes. Section 3 focuses on the method proposed to detect lava tubes. Section 4 is devoted to the description of the dataset and the experimental results. Finally, Section 5 presents the conclusions and the directions for future work.

2. LAVA TUBE MODEL

Lava tubes are tunnel-like structures developed in volcanic periods as a flow of lava moving on the surface. The difference of temperature between the lava and the outside environment generates a crust, which becomes a thick rooftop. Over time, tubes get empty, and they are believed hollow. Blair et al. [6] studied the stability of lunar lava tubes by varying the stress state, the roof thickness, and the tube sizes (fig. 1). The study outlined the possible presence of huge stable tubes (several km) with a thin rooftop (hundreds of m).

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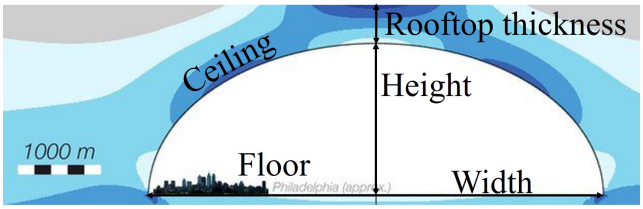


Fig. 1. A theoretical large dimension lava tube of 5 Km containing the city of Philadelphia. [Credit: Blair et al. [6]]

The approach proposed in this paper is based on the peculiar electromagnetic signature of lava tubes in radar sounder data, which is due to the transition between the materials encountered by the wave while penetrating the subsurface. In the following, the electromagnetic model of lava tube is described for both amplitude and phase, since radargrams are generated from a coherent signal. In radargrams, a lava tube is modeled as a complex target, since multiple linear features should co-exist showing specific relationships.

Lava tube amplitude model. The amplitude is expected to show three strong reflections that come from: the surface, the ceiling and the floor of the tube. The main reflection usually corresponds to the ground surface (G) and should be ideally a bright line as long as the radargram. Its amplitude is the strongest, since the surface is the first interface encountered by the wave, and so the incident power is almost equal the power transmitted by the sounder antenna. The pattern of a buried channel consists of a couple of isolated reflections that represent the tube interfaces and are generated by the strong contrast of the dielectric properties of the rock and the void. The one closer to the ground surface in range is the ceiling (C) of the tube, while the other is the floor (F). Their length along azimuth (l_C, l_F) depends on the tube size as well as on the angle of intersection between the tube and the flight direction of the sounder. Concerning the latter parameter, there are two limit cases: i) the two directions are parallel, and ii) they are perpendicular. In the parallel case, the lava tube signature consists of three lines as long as the radargram in azimuth generated by: the surface, the ceiling and the floor of the tube (Fig. 2 above, right). In the perpendicular case, the reflections have a hyperbolic shape and a length in azimuth shorter than the radargram (Fig. 2 below, right). Smoothly moving from the latter to the former case, the reflections preserve the (almost) linear shape, but they are stretched until they become as long as the radargram in the azimuth direction.

Lava tube phase model. In the phase domain, lava tube model considers three reflections having the same length and depth (i.e., range position) as the amplitude ones, but characterized by a phase inversion at each interface. The three inversions are generated by: i) the wave propagating from the void above the surface with a reflective index η_{void} to the rock in the subsurface with a reflective index $\eta_{rock} < \eta_{void}$, ii) the propagation from the rocky rooftop to the void inside the lava tube, and iii) to the transition from the void of the tube to the rock below the floor.

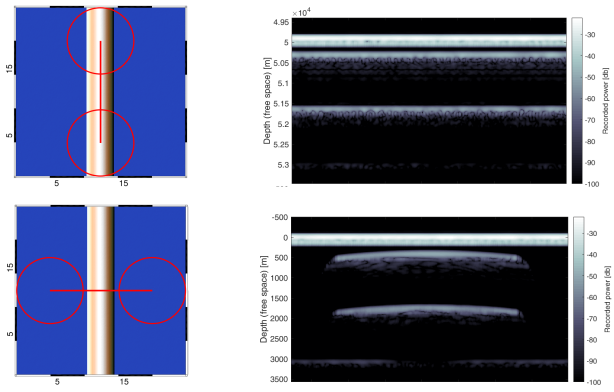


Fig. 2. Radar track and lava tube relative position (left) and simulated lava tube signature (right) for parallel (above) and perpendicular (below) cases.

3. PROPOSED APPROACH

Let us assume a clutter-free radargram where noise and clutter are previously mitigated. The proposed approach consists of two steps (Fig. 3): i) linear interfaces extraction from both the radargram amplitude and phase through a line-extraction algorithm, and i) tube detection by means of linear feature relationships analysis according to a fuzzy-logic-based system. The fuzzy rules analyze different aspects of the feature relationships by computing the membership values for each combination of them. Membership values are in the interval [0, 1] and indicate the degree to which the set of features represents a lava tube. The main membership function used here is a sigmoid as in eq. (1), where r is the parameter to test, c the inflection point, and a the slope.

$$\mu(r) = \frac{1}{1 + \exp\left(-a\left(r - c\right)\right)} \tag{1}$$

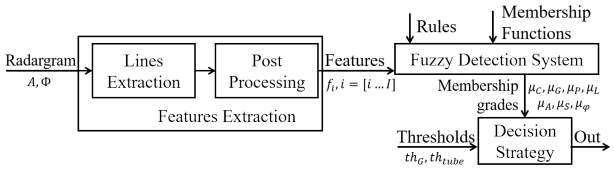


Fig. 3. Flow chart of the proposed approach.

3.1. Features Extraction

To extract the lava tube linear features in amplitude, a line extraction algorithm is used. Here, Carrer et al. [7] method is

adopted since it was developed for processing radar sounder data and manages both thermal and background noise.

The line extraction algorithm automatically detects layer boundaries (i.e., dielectrical interfaces) based on a local scale Hidden Markov Model (HMM) and the Viterbi Algorithm (VA). The approach consists of the following steps: layers enhancement, boundaries detection using a combination of VA and HMM, and edge combination. To enhance the layering structure of radargrams, an incoherent averaging filter is applied preserving the signal structure and reducing the noise. To further reduce the noise, signal free regions are detected by estimating the statistical properties of noise. Those are used to discriminate between samples associated with signal and noise. Then, layer continuity is enhanced by the convolution of radargrams with a low pass filter. In the second step, HMM and VA are applied to small blocks of radargrams to detect the layer boundaries by exploiting interference properties. HMM transforms the radargram into a trellis, which is then analyzed by the VA to search for the best path. Finally, once the best path for all the blocks is defined, the intensity of their pixels is thresholded to check if they identify layer boundaries. The algorithm stops when the thresholding condition is no longer verified, or it reaches the end of the radargram.

The line extraction algorithm generates a set of fragmented lines following the layer boundaries in the radargram. A post-processing is applied to cluster the segments belonging to the same boundary to create a set of candidate features $\mathcal{F} = \{f_i, i = 1 \dots I\}$. Each feature f_i is characterized by a length l_i, a range position ρ_i, an amplitude a_i, and a phase ϕ_i.

3.2. Proposed Fuzzy Detection Approach

The detection system for lava tubes is based on a set of fuzzy rules that analyze the candidate features $f_i \in \mathcal{F}$ detected by the line extraction algorithm. The fuzzy system evaluates the features to identify the best candidate to represent the ground surface (G), the ceiling (C) and the floor (F) of the tube. In the following, these rules are described in detail.

Completeness: verifies the presence of at least three features (i.e., $\mathcal{F} = \{f_i, i = 1 \dots I | I \geq 3\}$), which are the candidates to be the surface, the ceiling and the floor of the tube. The membership function associated is binary ($\mu_C = [0, 1]$).

Surface Detection: identifies the best candidate for the ground surface by evaluating the length of the feature. Let $r_G = l_i / l_R$ be the ratio between the length l_i of the candidate ground feature and the radargram length l_R. The membership function μ_G is evaluated by applying (1) to r_G.

Proportionality: verifies that the candidate features of the ceiling and the floor share azimuth coordinates. The proportionality membership μ_P is evaluated by (1) with input $r_P = min[L/l_i,\ L/l_j](i, j = 1 \dots I, i \neq j)$. L is the number of pixels with the same azimuth coordinate in the candidate features, l_i and l_j are the length of the candidate ceiling and floor feature, respectively.

Equivalence of Lengths: analyzes the lengths of the candidate features for the tube ceiling l_i and the floor l_j, which should be similar. The equivalence of lengths membership μ_L is evaluated by (1) with input $r_L = min[l_i/l_j,\ l_j/l_i],\ i \neq j$.

Alignment: analyzes both the parallelism among all the candidate features and their perpendicularity to the range direction. The centroids of the three candidate features are expected to lay on the same line, which should be parallel to the range direction. The associated membership function μ_A is evaluated by (1).

Amplitude Strength: analyzes the strength of the candidate reflections to check that the amplitude of the surface candidate is greater than the one of the candidate features for the tube ceiling and floor. Moreover, the strength of the candidate features for the ceiling and floor should be similar to each other, since losses inside the tube are almost null. The amplitude strength membership μ_S is evaluated by (1) with input $r_S = min[a_i/a_j,\ a_j/a_i](i, j = 1 \dots I, i \neq j)$. a_i and a_j are the mean amplitude of the candidate ceiling and floor features, respectively.

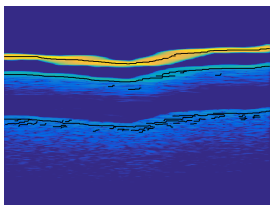
Phase Inversion: checks the presence of the phase inversion at each interface. Thus, for each couple of candidate features, the mean value ratio for the phase is evaluated by (1) resulting in μ_ϕ.

The best candidate to be the ground surface feature f_G is selected considering the Surface Detection rule and the depth in range of the candidate feature. Candidates are sorted according to their range coordinate and the first candidate from above with μ_G greater than a threshold (th_G) is selected as f_G. All the remaining features are analyzed in pairs to detect the possible presence of lava tubes. For each couple of candidate features (except the one identified as being f_G), the overall membership value μ is computed according to Larsen product implication (2). Finally, the presence or the absence of lava tubes is detected according to whether (μ) is greater than a threshold (th_{tube}) or not. Both the thresholds vary in the range [0, 1] and are defined experimentally.

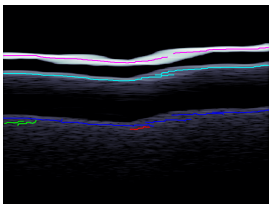
$$\mu = \mu_C\, \mu_P\, \mu_L\, \mu_A\, \mu_S\, \mu_\phi \tag{2}$$

4. EXPERIMENTAL RESULTS

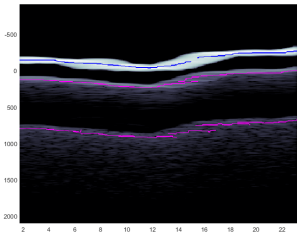
This approach was implemented and validated on simulated radargrams. Two kinds of experiments were conducted: the first by varying the dimensions and the depth of the tubes according to Blair et al. [6], the second by varying the characteristics of the terrain. In the first one, the radius of the tube varies from 250 m up to 2 km, considering small, medium and huge cases. The thickness of the tube rooftop varies accordingly, increasing up to 200 m. For all simulations both parallel and perpendicular cases are considered. For the terrain analysis, two different situations are taken into account considering as reference the case of the smooth surface. The first simulated terrain shows roughness. Thus, the surface



(b) Lines Extracted

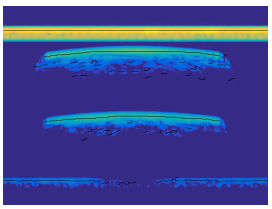


(b) Post-Processing Step

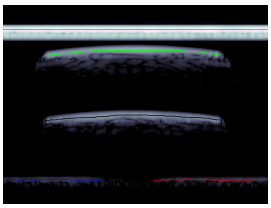


(c) Lava Tube Detected

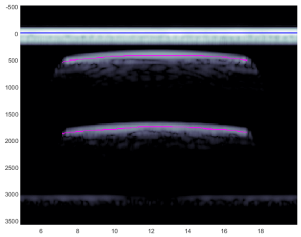
Fig. 4. Detection process for a radargram imaging a tube parallel to the sounder flight direction and a rough terrain.



(a) Line Extracted



(b) Post-Processing Step



(c) Fuzzy Detection System

Fig. 5. Detection process for a radargram imaging a tube perpendicular to the sounder flight direction and a smooth terrain.

line is affected by the reflected echoes scattered in all the directions due to the roughness effect. The second simulated terrain shows both roughness and thick crater that generates additional linear features right above and below the surface feature.

Fig. 4 shows the three steps of the approach in the case in which the direction of the tube is parallel to the flight direction of the sounder and the terrain is rough. Fig. 4.a shows the segments extracted by the line extraction algorithm, fig. 4.b the candidate features, and fig. 4.c shows the detected lava tube in magenta and the ground surface in blue. Fig. 5 reports the three steps of the approach in the case the tube is perpendicular to the flight direction of the sounder and the terrain is smooth. Fig. 5.a shows the segments extracted by the line extraction algorithm, fig. 5.b the candidate features, and fig. 5.c shows the lava tube detected in magenta and the ground surface in blue.

5. CONCLUSION

This paper proposes a model for the lava tube signature in radargrams derived according to the peculiar electromagnetic signature of these tunnels. Based on the model, an automatic approach for lava tube detection in radargrams is designed and implemented. The approach was validated by analyzing simulated radargrams showing lava tubes with different sizes and depth and different terrain characteristics in terms of roughness and presence of craters. The system demonstrated to be effective in all the cases and thus robust in absence of clutter. Preliminary experiments were conducted on real radargrams acquired by Kaguya mission (SELENE/JAXA) that look promising. As a future work, we plan to model other complex targets and to automatically extract the features by using deep neural network, such as convolutional neural networks.

6. REFERENCES

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